Lemma：F是域且 $\mathrm{char}(F)=0$，对$\forall n\in\mathbb{Z}\setminus\{0\}$和$\forall x\in F$，存在唯一的$y\in F$使得 $ny=x$．记 $y:=\frac{1}{n}x$．

Proof：对$\forall n\in\mathbb{Z}\setminus\{0\}$和$\forall x\in F$，

$\because$ F是一个域 $\therefore$ 存在唯一的环同态 $\mathbb{Z}\to F$．

$\eta_F:\mathbb{Z}\to F$, $x\mapsto x\cdot 1_F$ 就是 $\mathbb{Z}\to F$的唯一的环同态．

$\because$ $n\in\mathbb{Z}\setminus\{0\}$ $\therefore$ $n\neq 0$ $\because$ $\eta_F:\mathbb{Z}\to F$, $x\mapsto x\cdot 1_F$ 是单同态 (3.7(4)笔记)

$\therefore$ $\eta_F(n)\neq\eta_F(0)$ $\therefore$ $n\cdot 1_F\neq 0_F$ $\therefore$ $n\cdot 1_F\in F\setminus\{0_F\}=F^\times$

$\therefore$ $(n\cdot 1_F)^{-1}\in F$．令 $y=(n\cdot 1_F)^{-1}\cdot x$ $\therefore$ $y\in F$

$\because$ $ny=(n\cdot 1_F)y=(n\cdot 1_F)(n\cdot 1_F)^{-1}x=1_F x=x$．存在性得证．

假设存在 $y_1,y_2\in F$，s.t. $n\cdot y_1=x$ 且 $n\cdot y_2=x$

$\therefore$ $n\cdot y_1=n\cdot y_2$ $\therefore$ $(n\cdot 1_F)y_1=(n\cdot 1_F)y_2$

$\because$ $n\neq 0$ $\therefore$ $\eta_F(n)\neq\eta_F(0)$ $\therefore$ $n\cdot 1_F\neq 0_F$ $\therefore$ $n\cdot 1_F\in F^\times$

$\therefore$ $(n\cdot 1_F)^{-1}(n\cdot 1_F)y_1=(n\cdot 1_F)^{-1}(n\cdot 1_F)y_2$ $\therefore$ $y_1=y_2$ 唯一性得证．

记 $y=(n\cdot 1_F)^{-1}\cdot x:=\frac{1}{n}x$． $\square$

Lemma：F是域且 $\mathrm{char}(F)=p$，p是素数．对$\forall n\in\mathbb{Z}$且有 $p\nmid n$，$\forall x\in F$，存在唯一的$y\in F$使得 $ny=x$．记 $y=\frac{1}{n}x$．

Proof: 对 $\forall n\in\mathbb{Z}$ 且有 $p\nmid n$，$\forall x\in F$，

假设 $n\cdot 1_F=0_F$，则有：$\because n\in\mathbb{Z}$，$p\in\mathbb{Z}_{\geq 2}$，$p\neq 0$

$\therefore$ 存在唯一的 $q,r\in\mathbb{Z}$，s.t. $n=pq+r$，$0\leq r<p$

$\because p\nmid n$ $\quad\therefore r\neq 0$ $\quad\therefore 0<r<p$

$\because n\cdot 1_F=0_F$ $\quad\therefore (pq+r)\cdot 1_F=0_F$

$\therefore 0_F=n\cdot 1_F=(pq+r)\cdot 1_F=(pq)\cdot 1_F+r\cdot 1_F=(qp)\cdot 1_F+r\cdot 1_F$

$=q(p\cdot 1_F)+r\cdot 1_F=q\cdot 0_F+r\cdot 1_F=0_F+r\cdot 1_F=r\cdot 1_F$

$\therefore r\cdot 1_F=0_F$ $\quad\therefore \mathrm{char}(F)\mid r$ $\quad\therefore p\mid r$ 矛盾.

$\therefore n\cdot 1_F\neq 0_F$ $\quad\therefore n\cdot 1_F\in F\setminus\{0_F\}=F^{\times}$ $\quad\therefore (n\cdot 1_F)^{-1}\in F$

令 $y=(n\cdot 1_F)^{-1}x$ $\quad\therefore y\in F$

$\therefore ny=(n\cdot 1_F)y=(n\cdot 1_F)(n\cdot 1_F)^{-1}x=x$ 存在性得证.

假设存在 $y_1,y_2\in F$，使得 $n\cdot y_1=x$ 且 $n\cdot y_2=x$

$\therefore n\cdot y_1=n\cdot y_2$ $\quad\therefore (n\cdot 1_F)y_1=(n\cdot 1_F)y_2$

$\because n\cdot 1_F\in F^{\times}$ $\quad\therefore (n\cdot 1_F)^{-1}(n\cdot 1_F)y_1=(n\cdot 1_F)^{-1}(n\cdot 1_F)y_2$

$\therefore y_1=y_2$. 唯一性得证.

记 $y=(n\cdot 1_F)^{-1}x:=\frac{1}{n}x$ $\quad\square$

定理（Wilson's theorem）p是素数，则有：$(p-1)! \equiv -1 \pmod{p}$

Proof：当 $p=2$ 时，$(2-1)!-(-1)=1+1=2$，$2|2$

$\therefore (2-1)! \equiv -1 \pmod{2}$

当 $p=3$ 时，$(3-1)!-(-1)=2+1=3$，$3|3$

$\therefore (3-1)! \equiv -1 \pmod{p}$ $\quad \therefore (3-1)! \equiv -1 \pmod{3}$

当 $p=5$ 时，$(5-1)!-(-1)=24+1=25$，$5|25$

$\therefore (5-1)! \equiv -1 \pmod{5}$

当 $p=7$ 时，$(7-1)!-(-1)=720+1=721$，$7|721$

$\therefore (7-1)! \equiv -1 \pmod{7}$

当 $p=11$ 时，$(11-1)!-(-1)=362880|$，$11|362880|$

$\therefore (11-1)! \equiv -1 \pmod{11}$

对 $\forall p \in \mathbb{Z}_{\geq 11}$ 且 p 是素数，有：

$\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} = \{[x]_p \mid x \in \mathbb{Z}\} = \{[0]_p, [1]_p, [2]_p, \cdots, [p-1]_p\}$ 是域.

$\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} = \{x+p\mathbb{Z} \mid x \in \mathbb{Z}\}$，$\mathbb{F}_p^{\times} = \{[1]_p, [2]_p, \cdots, [p-1]_p\}$

对 $\forall \lambda \in \mathbb{F}_p^{\times}$，

若 $\lambda = \lambda^{-1}$，则 $\lambda^2 = \lambda \cdot \lambda = \lambda \cdot \lambda^{-1} = 1_{\mathbb{F}_p} = 1_{\mathbb{Z}/p\mathbb{Z}} = 1+p\mathbb{Z} = [1]_p$

$\therefore (\lambda+[1]_p)(\lambda-[1]_p) = \lambda(\lambda-[1]_p) + [1]_p(\lambda-[1]_p)$

$= \lambda^2 - \lambda \cdot [1]_p + [1]_p \cdot \lambda - [1]_p \cdot [1]_p = \lambda^2 - [1]_p = [1]_p - [1]_p = [0]_p$

$\therefore \lambda+[1]_p \in \mathbb{F}_p$，$\lambda-[1]_p \in \mathbb{F}_p$，$(\lambda+[1]_p)(\lambda-[1]_p) = 0_{\mathbb{F}_p}$

$\because \mathbb{F}_p$是域 $\therefore \mathbb{F}_p$是整环

$\therefore \lambda + [1]_p = 0_{\mathbb{F}_p}$ 或 $\lambda - [1]_p = 0_{\mathbb{F}_p}$

$\therefore \lambda = [-1]_p = [p-1]_p$ 或 $\lambda = [1]_p$

若 $\lambda = [-1]_p$，则 $\lambda \cdot \lambda = \lambda^2 = [-1]_p^2 = [(-1)^2]_p = [1]_p$ $\therefore \lambda = \lambda^{-1}$

若 $\lambda = [1]_p$，则 $\lambda \cdot \lambda = \lambda^2 = [1]_p^2 = [1]_p$ $\therefore \lambda = \lambda^{-1}$

$\therefore$ 对 $\forall \lambda \in \mathbb{F}_p^\times$，$\lambda = \lambda^{-1} \Longleftrightarrow \lambda = [-1]_p$ 或 $\lambda = [1]_p$

$\therefore \mathbb{F}_p^\times$中所有元素的乘积 $= [1]_p \cdot [2]_p \cdot \cdots \cdot [p-1]_p = [(p-1)!]_p$

$\mathbb{F}_p^\times$中所有元素的乘积 $= [1]_p \cdot [p-1]_p = [p-1]_p = [-1]_p$

$\therefore [(p-1)!]_p = [-1]_p$ $\therefore (p-1)! \equiv -1 \pmod p$ $\square$

定理（Wilson's theorem）~~$x \in \mathbb{Z}$~~ $\alpha \in \mathbb{Z}_{\geq 2}$，$(\alpha-1)! \equiv -1 \pmod \alpha$，则有：$\alpha$是素数.

Proof：假设α不是素数，则有：$\alpha \in \mathbb{Z}_{\geq 2}$且$\alpha$不是素数.

$\therefore \exists d \in \mathbb{Z}$ 且 $1 < d < \alpha$，s.t. $d \mid \alpha$

$\therefore \exists k \in \mathbb{Z}$，s.t. $\alpha = dk$

$\because d \in \mathbb{Z}$ 且 $1 < d < \alpha$ $\therefore d \in \mathbb{Z}$ 且 $2 \leq d \leq \alpha - 1$

$\therefore d \mid (\alpha-1)!$ $\therefore (\alpha-1)! \equiv 0 \pmod d$

$\because (\alpha-1)! \equiv -1 \pmod \alpha$ $\therefore \alpha \mid ((\alpha-1)! - (-1))$

$\therefore \exists \beta \in \mathbb{Z}$，s.t. $(\alpha-1)! - (-1) = \alpha\beta$

$\therefore (\alpha-1)!-(-1)=\alpha\beta=(\alpha k)\beta=\alpha(k\beta)$

$\because (\alpha-1)!-(-1)=\alpha(k\beta)$，$k\beta\in\mathbb{Z}$，$\alpha\in\mathbb{Z}$且$2\leqslant\alpha\leqslant\alpha-1$

$\therefore \alpha \mid (\alpha-1)!-(-1)$　　$\therefore (\alpha-1)!\equiv -1 \pmod{\alpha}$

$\therefore 0\equiv -1 \pmod{\alpha}$　　$\therefore \alpha \mid 0-(-1)$　　$\therefore \alpha \mid 1$　　$\therefore \alpha=\pm 1$

矛盾.　$\therefore$ α是素数　$\square$

定理(Wilson's theorem) $p\in\mathbb{Z}_{\geqslant 2}$，则有：

p是素数 $\Longleftrightarrow (p-1)!\equiv -1 \pmod{p}$

Proof: 上两个定理已证. $\square$

Lemma: R是环，对$\forall r\in R$，有：$r^2=r$，则有：R是交换环.

Proof: 对$\forall x\in R$，$\because R$是环　$\therefore x+x\in R$　$\therefore (x+x)^2=x+x$

$$\begin{aligned}\therefore x+x &= (x+x)^2=(x+x)(x+x)=x(x+x)+x(x+x)\\ &= x^2+x^2+x^2+x^2 = x+x+x+x\end{aligned}$$

$$\begin{aligned}\therefore x+x &= (x+x)+0_R = (x+x)+\big((x+x)+(-(x+x))\big)\\ &= \big((x+x)+(x+x)\big)+\big(-(x+x)\big)\\ &= (x+x+x+x)+\big(-(x+x)\big)\\ &= (x+x)+\big(-(x+x)\big)=0_R\end{aligned}$$

$\therefore$对$\forall x\in R$，$x+x=0_R$

$\because$ 对 $\forall x \in R$, $x = x + 0_R = x + (x + (-x)) = (x+x) + (-x)$

$= 0_R + (-x) = -x$

$\therefore$ 对 $\forall x \in R$, $x = -x$.

对 $\forall x, y \in R$, $\because R$ 是环 $\therefore x+y \in R$ $\therefore (x+y)^2 = x+y$

$\therefore x+y = (x+y)^2 = (x+y)(x+y) = x(x+y) + y(x+y)$

$= x \cdot x + xy + yx + y \cdot y = x^2 + xy + yx + y^2$

$= x + xy + yx + y = (x+y) + (xy+yx)$

$\therefore xy + yx = 0_R + (xy+yx) = \big((-(x+y)) + (x+y)\big) + (xy+yx)$

$= (-(x+y)) + \big((x+y) + (xy+yx)\big)$

$= (-(x+y)) + (x+y) = 0_R$

$\therefore xy = xy + 0_R = xy + (yx + (-yx)) = (xy+yx) + (-yx)$

$= 0_R + (-yx) = -yx = yx$

$\therefore R$ 是交换环. $\square$

定义(布尔环) R 是环, 若对 $\forall r \in R$, 有: $r^2 = r$, 则称环 R 是布尔环.

Lemma: 布尔环是交换环.

Proof: 上面已证. $\square$